INTRODUCTION

In our day-today life, we are supposed to use many electrical appliances such as water pump, iron box, mixie, grinder, washing machine, UPS, TV etc. In order to understand the basic working principle of these appliances, it is better to know about some of the important electrical devices which are used in the appliances.

In this, we are going to read about

1. Cells
2. Inductors
3. Transformers
4. Switches and Fuses

2.1. CELLS

A device which converts chemical energy into electrical energy is called as a cell. A group of cells connected together is called battery.

In electric and electronic circuits, an electrical cell is denoted by the following symbol as shown in Figure 2.1.

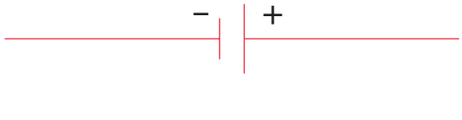


Figure 2.1 Symbol for Cell

When two dissimilar metals or electrodes are immersed in an electrolyte, there will be a potential difference produced between these metals or electrodes. In this, one electrode acts as positive terminal and another electrode acts as negative terminal.

History



In 1800, "Alessandro Volta" invented the 'Voltaic Pile' and discovered the first practical method of generating electricity, as we already discussed in the Chapter 1. This 'Voltaic pile' was the first 'wet cell' that produced a reliable, steady current of electricity.

2.1.1. Types of Cells

The cells are classified into two categories. They are,

1. Primary cells
2. Secondary cells

Primary Cells

Primary cells are widely used in clocks, watches, remote controls and calculators.

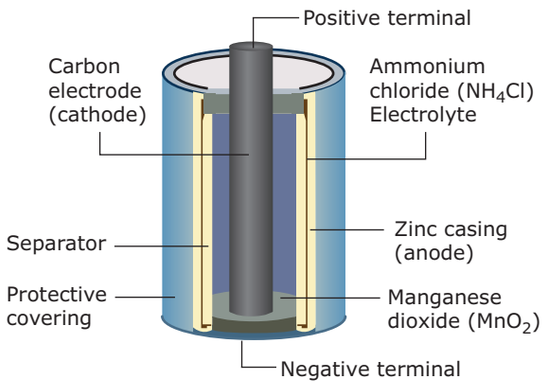


Figure 2.2(a) Structure of Primary Cell

The cells that cannot be recharged are called as 'primary cells'. Some of the primary cells are Alkaline cell, Lithium cell, Zinc-Carbon cell or dry cell.

Generally, primary cells (figure 2.2a) are available in the voltage rating of 1.5 V, 3 V, 9 V and 12 V and the current rating of 0.2 A, 0.5 A, 1 A and 2 A. AA and AAA

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